

Question 1 of 8

1 Points

Let $f(n) = 3n^2 + 8n - 4$ and $g(n) = 8n^2 + 3n + 4$. Then which of the following is wrong?

- ☐ A. $f(n)$ is $O(g(n))$
- ☐ B. $g(n)$ is $O(f(n))$
- ☐ C. $f(n) - g(n)$ is $O(1)$
- ☐ D. $f(n) + g(n)$ is $O(n^2)$

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Question 2 of 8

1 Points

Let $f(n) = 10n^2 + 19$ and $g(n) = 25n^2 + 100n$.

- ☐ A. $f(n) > g(n)$ for all $n > 0$ and $f(n)$ is $O(g(n))$
- ☐ B. $f(n)$ is not $O(g(n))$. But $g(n)$ is $O(f(n))$.
- ☐ C. $f(n)$ is not $O(g(n))$ and $g(n)$ is not $O(f(n))$.
- ☐ D. $f(n)$ is $O(g(n))$ and $g(n)$ is $O(f(n))$.

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Let $T(n)$ denote the time required to perform binary search on an array of size n . Then

- ☐ A. $T(n) = 2T(n/2) + c$ and $T(1) = d$ for some constants c and d .
- ☐ B. $T(n) = 2T(n/2) + cn$ and $T(1) = d$ for some constants c and d .
- ☐ C. $T(n) = T(n/2) + cn$ and $T(1) = d$ for some constants c and d .
- ☐ D. None of the above.

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Let $T(n)$ denote the time required to perform merge sort on an array of size n . Then

- ☐ A. $T(n) = 2T(n/2) + c$ and $T(1) = d$ for some constants c and d .
- ☐ B. $T(n) = 2T(n/2) + cn$ and $T(1) = d$ for some constants c and d .
- ☐ C. $T(n) = T(n/2) + cn$ and $T(1) = d$ for some constants c and d .
- ☐ D. None of the above

Question 5 of 8 1 Points

You have divided the problem into 5 subproblems each one of them is of the size $n/3$. If there is no extra work, what will be the recurrence relation for the time complexity $T(n)$?

- ☐ A. $T(n) = 3T(n/5) + 5n/3$, $T(1) = d$
- ☐ B. $T(n) = 5T(n/3) + c$, $T(1) = d$
- ☐ C. $T(n) = 3T(n/5) + c$, $T(1) = d$
- ☐ D. $T(n) = 5T(n/3) + cn$, $T(1) = d$

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Question 8 of 8 1 Points

What is the "Width" and "Depth" of merge sort algorithm as explained by Dr. Nair

- ☐ A. Width = n , Depth = $\log n$
- ☐ B. Width = 2, Depth = $\log n$
- ☐ C. Width = $\log n$, Depth = n
- ☐ D. None of the above.

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